1 - Cyclic Loading (Step 1)

mohr(σxmax,σxmin,σymax,σymin,τmax,τmin)

σ_mean,x		σ_alt,x	
σ_mean,y		σ_alt,y	
τ_mean,xy		τ_alt,xy	

$$\sigma/\tau_{mean,x/y/z} = \frac{\sigma/\tau_{max} + \sigma/\tau_{min}}{2}$$

$$\sigma/\tau_{alt,x/y/z} = \frac{\sigma/\tau_{max} - \sigma/\tau_{min}}{2}$$

2 - Mohr Circle (Step 2)

$$R_{mean}: \sqrt{\left(\frac{\sigma_{mean,x} - \sigma_{mean,y}}{2}\right)^2 + \left(\tau_{mean,xy}\right)^2}$$

$$\sigma_{mean,center} = \frac{\sigma_{mean,x} + \sigma_{mean,y}}{2}$$

$$\sigma_{mean,1} = \sigma_{mean,center} + R_{mean}$$

$$\sigma_{mean,3} = \sigma_{mean,avg} - R_{mean}$$

$$R_{alt}: \sqrt{\left(\frac{\sigma_{alt,x} - \sigma_{alt,y}}{2}\right)^2 + \left(\tau_{alt,xy}\right)^2}$$

$$\sigma_{alt,1} > \sigma_{alt,2} > \sigma_{alt,3}$$

3 - Stress Theories (Step 3, Choose 1)

Maximum Shear Stress Theory (MSST) (Tresca):

MSST Effective Stress $= \sigma' = \sigma_1 - \sigma_3$

If all tension, safe when $N = \frac{S_y}{\sigma_1} > 1$

If all compression, safe when $N = \frac{S_y}{|\sigma_3|} > 1$

If t and c, safe when $N = \frac{S_y}{\sigma'} > 1$

Distortion Energy Theory (DET) (von Mises):

$$\sigma_e > S_y \quad = Fail$$

$$\sigma_e: \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$N = \frac{S_y}{\sigma_e}$$

Coulomb Mohr Theory (CMT):

If Tension and Compression, Safe when N:

$$\frac{\sigma_1}{S_{ut}} + \frac{|\sigma_3|}{|S_{uc}|} < \frac{1}{N}$$

If All in Tension: $N = \frac{S_{ut}}{\sigma_1} > 1$

If All in Compression: $N = \frac{|S_{uc}|}{|\sigma_3|} > 1$

4 - Fatigue Failure (Step 4, Choose 1)

Soderberg (No Deformation): $\frac{K_f \sigma_{alt}}{S_n'} + \frac{\sigma_m}{S_y} = \frac{1}{N}$

* Use when Sy and S'n are given

Goodman (Linear): $\frac{\sigma_{eq,alt}}{S_e} + \frac{\sigma_{eq,mean}}{S_{ut}} = \frac{1}{N}$

Gerber (Parabolic): $\frac{K_f N \sigma_a}{S_n'} + \left(\frac{N\sigma_m}{S_{ut}}\right)^2 = 1$

5 - Pressure Vessel Cyclic Load

Wall Thickness Check:

$$\frac{r}{t} > 10, \text{is thin wall}$$

$$\frac{r}{t} < 10, \text{is thick wall}$$

Thick-Walled (Lame) Equation:

lame_ext(ID,OD,P,r_pressure) @ pi=0
lame_intl(ID,OD,P,r_pressure) @ po = 0

$$\sigma_1 = \sigma_{\text{hoop}}, \sigma_2 = \sigma_{\text{axial}}, \sigma_3 = 0$$

When p_o = 0 (internal pressure):

Axial Stress: $\sigma_a = \frac{p_i r_i^2}{r_o^2 - r_i^2}$

Hoop Stress: $\sigma_\theta = \frac{p_i r_i^2}{r_o^2 - r_i^2}\left(1 + \frac{r_o^2}{r^2}\right)$

Radial Stress: $\sigma_r = \frac{p_i r_i^2}{r_o^2 - r_i^2}\left(1 - \frac{r_o^2}{r^2}\right)$

When p_i = 0 (external pressure):

$$\sigma_a = \frac{-p_o r_o^2}{r_o^2 - r_i^2}$$

$$\sigma_\theta = \frac{-p_o r_o^2}{r_o^2 - r_i^2}\left(1 + \frac{r_o^2}{r^2}\right)$$

$$\sigma_r = \frac{-p_o r_o^2}{r_o^2 - r_i^2}\left(1 - \frac{r_o^2}{r^2}\right)$$

When there is pi and po:

$$\sigma_a = \frac{p_i r_i^2 - p_o r_o^2}{r_{o^2} - r_i^2} = \frac{F}{A}$$

$$\sigma_\theta = \frac{p_i r_i^2 - p_o r_o^2}{r_{o^2} - r_i^2} + \frac{r_i^2 r_o^2 (p_i - p_o)}{r^2 (r_o^2 - r_i^2)}$$

$$\sigma_r = \frac{p_i r_i^2 - p_o r_o^2}{r_{o^2} - r_i^2} - \frac{r_i^2 r_o^2 (p_i - p_o)}{r^2 (r_o^2 - r_i^2)}$$

Thin Wall Equation:

$$\sigma_1 = \sigma_{\text{hoop}} = \frac{Pr}{t}$$

$$\sigma_2 = \sigma_{\text{axial}} = \frac{Pr}{2t}$$

$$\sigma_3 = 0$$

$$N = \frac{S_y}{\sigma_1}$$

6 - Interference Fits

$$\delta = \alpha \times \Delta T \times L$$

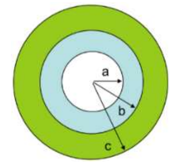
a = coefficient of thermal expansion, dT = temperature change, L = nominal dimension

Use max interference for stress calcs!

1. Determine interference δ
2. Compute interface pressure

1-material p:

$$= \frac{E\delta}{2b}\left[\frac{(c^2 - b^2)(b^2 - a^2)}{2b^2(c^2 - a^2)}\right]$$



2-material p:

$$= \frac{\delta}{2b\left[\frac{1}{E_o}\left(\frac{c^2 + b^2}{c^2 - b^2}\right) + \frac{1}{E_i}\left(\frac{b^2 + a^2}{b^2 - a^2} - v_i\right)\right]}$$

Material	E (psi)	v
Steel	30 x 10^6	0.27
Aluminum	10 x 10^6	0.31
Copper	17 x 10^6	0.34
Brass	14 x 10^6	0.34
Bronze	17 x 10^6	0.27

3. Compute Stresses

Stress in the Outside Member:

$$\sigma_o = p\left(\frac{c^2 + b^2}{c^2 - b^2}\right)$$

Stress in the Inner Member:

$$\sigma_i = -p\left(\frac{b^2 + a^2}{b^2 - a^2}\right)$$

7 - Combined Cyclic Loading

1. Calculate I = ___, J = ____, Area = ___ and Q = y_bar*A

Cross Section	Ix	Iy	J
Solid Circle	$\frac{\pi r^4}{4} = \frac{\pi d^4}{64}$	same	2I
Thick Tube	$\frac{\pi}{4}(R_o^4 - R_i^4)$	same	2I
Solid Square	$\frac{a^4}{12}$	same	$\frac{a^4}{6}$
Rectangle	$\frac{bh^3}{12}$	$\frac{hb^3}{12}$	$\frac{bh}{12}(b^2 + h^2)$

2. List out Forces and Draw Cross Section

Force	Min	Max
Fx		
Fy		
Fz		
Mx		
My		
Mz		

3. List out Min and Max Stresses

Stress	Min	Max
σ_x		
σ_y		
τ_{xy}		

Causes of σ: $\frac{Pr}{t}, \frac{Pr}{2t}, \frac{N}{A}, \frac{Mc}{I}$

Causes of τ: $\frac{VQ}{It}$ *(Check for zero Q bc of ⊥ V)*, $\frac{Tc}{J}$

4. Solve using **1-Cyclic Loading**

8.1 – Normal Column Design and Analysis

Condition	K (theoretical)	K (practical)
Pinned-Pinned	1.0	1.0
Fixed-Fixed	0.5	0.65
Fixed-Free	2.0	2.10
Fixed-Pinned	0.7	0.8

1.1 Calculate I = __ (refer to 7 – CCL)
note: if rectangle I, use smaller value

1.2 Calculate/Find A = __
1.3 List out K = __
1.4 Calculate $L_{eff} = KL =$ __

2. Calculate Radius of Gyration: $r = \sqrt{\frac{I}{A}}$

3. Calculate Slenderness Ratio: $S_r = \frac{L_{eff}}{r}$

4. Calculate Column Constant: $C_c = \sqrt{\frac{2\pi^2 E}{S_y}}$

5. Column Length Check:

$S_r > C_c \rightarrow$ long column

$S_r < C_c \rightarrow$ short column

6.1. **Short Column (Use Johnson's):**

$$\sigma_{cr-J} = S_y\left[1 - \frac{S_y S_r^2}{4\pi^2 E}\right]$$

$$P_{cr} = AS_y\left[1 - \frac{S_y S_r^2}{4\pi^2 E}\right]$$

Note That: $S_r = C_c \Leftrightarrow$ Johnson = Euler

Note That: $S_r = 0 \Leftrightarrow \frac{P_{cr}}{A} = S_y$

6.2. **Long Column (Use Euler's):**
Critical Load (First Buckling Mode):

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

In Terms of Slenderness Ratio:

$$P_{cr} = \frac{\pi^2 EA}{S_r^2}$$

8.2 – Crooked Column Analysis (Only Applies to Long Column, Use Euler)

1. Do Steps 1-4 from 8.1

2. List out a = ___ (crookedness)

3. Calculate c = D/2 = __

4. Calculate C1 using:

$$C_1 = -\frac{1}{N}\left(S_y A + \left(1 + \frac{a\ (crookedness) * c}{r^2\ (radius\ of\ gyration)}\right)P_{Cr}\right)$$

5. Calculate C2 using:

$$C_2 = \frac{S_y A P_{cr}}{N^2}$$

6. Solve for Pa using polyRoot(x^2+, x), use lowest positive number:

$$P_a^2 + C_1 P_a + C_2 = 0$$

7.1 Compare Pa with Pcr at Safety Factor N:

$$P_{cr} = \frac{P_{\text{Cr (Euler)}}}{N}$$

7.2 Determine change in critical load:

Use $P_{\text{cr (Euler)}}$ at N = 1 and N = #

8.4 – Crooked Column Analysis (Only Applies to Long Column, Use Euler)

1. Do Steps 1-3 from 8.2
2. List out e = __
3. List out E = __
4. List out S_y = __
5. Calculate

$$\sigma_{L/2} = \sigma_{max} = \frac{P}{A}\left(1 + \frac{e\ (eccentricity) * c}{r^2} Sec\left(\frac{L_{eff}}{2r}\sqrt{\frac{P}{AE}}\right)(rad)\right)$$

9 – Other Equations

Endurance Strength

$S'_n = (S_u C_{sf})C_s C_R C_m C_{st} = S_n C_s C_R C_m C_{st}$
S_n – Surface Finish Factor
C_s – Size Factor
C_R – Reliability Factor
C_m – Material Factor
C_{st} – Stress Type Factor

Damage Accumulation

$$Damage = \frac{\#\ of\ Cycles\ Loaded}{\#\ of\ Cycles\ till\ Failure}$$

Units

giga-	G	10^9
mega-	M	10^6
kilo-	k	10^3
centi-	c	10^{-2}
milli-	m	10^{-3}
micro-	µ	10^{-6}
nano-	n	10^{-9}
lb / psi	x1000	kip / ksi

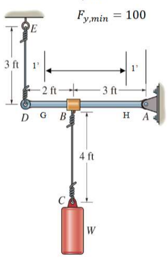
Chapter 5 – Failure Theories

Fatigue Failure Theories

	Static Loading	Cyclic Loading
Ductile Material	MSST, DET	Soderberg Goodman Gerber
Brittle Material	CMT, MMT	N/A (do not use brittle material for cyclic loading)

Cyclic Loading Example

Cyclic loading
The object (W = 500 lb) is moving between positions G and H back and forth repeatedly. The wire rope DE is made of steel with S_y = 80 ksi, S_n' = 20 ksi. Determine the required wire rope diameter based on Soderberg criterion and N = 2.



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From FBD, force acting on rope DE

$F_{y,max} = 400$

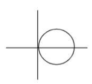
$F_{y,min} = 100$

$\sigma_{x,max} = 0$, $\sigma_{x,min} = 0$, $\sigma_{y,max} = \frac{400}{A}$, $\sigma_{y,min} = \frac{100}{A}$, $\tau_{xy,max} = 0$, $\tau_{xy,min} = 0$

$\sigma_{x,m} = 0$, $\sigma_{x,a} = 0$, $\sigma_{y,m} = \frac{250}{A}$, $\sigma_{y,a} = \frac{150}{A}$, $\tau_{xy,m} = 0$, $\tau_{xy,a} = 0$

Mean

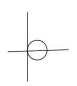
$\sigma_{x,m} = 0$, $\sigma_{y,m} = \frac{250}{A}$, $\tau_{xy,m} = 0$



$\sigma_{m1} = \frac{250}{A}$, $\sigma_{m2} = 0$, $\sigma_{m3} = 0$

Alternating

$\sigma_{x,a} = 0$, $\sigma_{y,a} = \frac{150}{A}$, $\tau_{xy,a} = 0$



$\sigma_{a1} = \frac{150}{A}$, $\sigma_{a2} = 0$, $\sigma_{a3} = 0$

MSST $\quad \sigma_m' = \frac{250}{A} \quad \sigma_a' = \frac{150}{A}$

Soderberg

$$\frac{1(\frac{150}{A})}{20{,}000} + \frac{\frac{250}{A}}{80000} = \frac{1}{2}$$

$A = 0.02125$

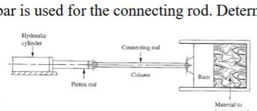
$D = 0.1645$

Check, $D < 1$"

assumption, $C_s = 0.88$ is OK

Column Analysis

- A 24" steel bar is used for the connecting rod. Determine the diameter if the force acting on the ram is 2000 lb.



$P_{cr} = \frac{\pi^2 EI}{L^2}$ Pinned-pinned, K = 1

Buckling evaluation

$$P_{cr} = \frac{\pi^2 EI}{L_{eff}^2} \qquad 2000 = \frac{\pi^2(30\times 10^6)I}{(24)^2} \qquad I = \frac{2000(24)^2}{\pi^2(30\times 10^6)} = 0.00389 \qquad \boxed{D = 0.53"}$$

Yielding evaluation

If $S_y = 30000$ psi $\qquad \sigma = \frac{2000}{A} < S_y \qquad A > \frac{2000}{30000} = 0.067 \qquad \boxed{D = 0.29"}$

Due to the buckling concern, the larger diameter (D = 0.53") is needed.

- Design a 12.75' length (fixed-free end) column (yield strength = 60 ksi) to support a 4000 lb load.

12.75' = 153" Assume long column Use Euler's equation Fixed-free, K = 2

$$P_{cr} = \frac{\pi^2 EI}{L_{eff}^2} \qquad 4000 = \frac{\pi^2(30\times 10^6)I}{(2\times 153)^2} \qquad I = \frac{4000(306)^2}{\pi^2(30\times 10^6)} = 1.26$$

$\boxed{\text{Select } 3\times 3\times 1/4}$

Check long column assumption (assume 60 ksi yield strength)

$$S_r = \frac{L_{eff}}{r} = \frac{306}{1.1} = 278 \qquad C_c = \sqrt{\frac{2\pi^2 E}{S_y}} = \sqrt{\frac{2\pi^2(30\times 10^6)}{60000}} = 99.34$$

$S_r > C_c \rightarrow$ long column

- A 48" (length) 8 x 4 x ½ steel structure tubing (1020 hot rolled) is used to support a 4000 lb load. The end condition is designed as fixed-free. Estimate the safety factor of the design.
- Can tubing 2 x 2 x ¼ steel tubing be used? $\qquad S_y = 30000$ psi

$8\times 4\times {}^1/_2$ $\qquad S_r = \frac{L_{eff}}{r} = \frac{96}{1.54} = 62.3 \qquad C_c = \sqrt{\frac{2\pi^2 E}{S_y}} = \sqrt{\frac{2\pi^2(30\times 10^6)}{30000}} = 140$

$S_r < C_c \rightarrow$ short column

$$P_{cr} = AS_y\left[1 - \frac{S_y S_r^2}{4\pi^2 E}\right] = (10.4)(30000)\left[1 - \frac{(30000)62.3^2}{4\pi^2(30\times 10^6)}\right] = 281326 \text{ lb} \qquad N = \frac{281326}{4000} = 70.3$$

$2\times 2\times {}^1/_4$ $\qquad S_r = \frac{L_{eff}}{r} = \frac{96}{0.694} = 138.3 \qquad C_c = 140 \qquad S_r < C_c \rightarrow$ short column

$$P_{cr} = AS_y\left[1 - \frac{S_y S_r^2}{4\pi^2 E}\right] = (1.59)(30000)\left[1 - \frac{(30000)138.3^2}{4\pi^2(30\times 10^6)}\right] = 24590 \text{ lb} \qquad N = \frac{24590}{4000} = 6.15$$

Safe

Crooked Columns Examples

- The end condition of a column is pinned – pinned. The column length is 32". The diameter is 0.75". The initial crookedness is 0.125". The material is SAE 1040 hot-rolled. Calculate the allowable load based on N = 3. $\quad$ Pinned-pinned, $K = 1$ $\quad S_y = 42000$ psi

$A = 0.442$

$I = 0.01553$

$r = \sqrt{I/A} = D/4 = 0.188$

$c = D/2 = 0.376$

$a = 0.125$

$S_r = \frac{32}{0.188} = 170.2 \qquad C_c = \sqrt{\frac{2\pi^2 E}{S_y}} = \sqrt{\frac{2\pi^2(30\times 10^6)}{42000}} = 118.7$

$S_r > C_c$, long column, Euler's Equation

$P_{cr} = \frac{\pi^2 EI}{L_{eff}^2} = \frac{\pi^2(30\times 10^6)(0.01533)}{32^2} = 4490$ lb $\qquad$ Note: N = 1

for $N = 3$, $P_{cr} = \frac{4490}{3} = 1497$

$P_a^2 + C_1 P_a + C_2 = 0$

$C_1 = -\frac{1}{N}\left[S_y A + \left(1 + \frac{ac}{r^2}\right)P_{cr}\right] = -9675$

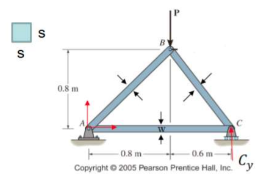
$C_2 = \frac{S_y A P_{cr}}{N^2} = 9260000$

$P_a^2 - 9675P_a + 9260000 = 0$

$P_a = 1076.7$

$P_a = 8572.3$

Note: N = 3

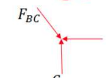


C_y

Determine the cross sections of AB and BC for P = 2000 N and Sy= 210 MPa

$\sum M_A = 0 \qquad 2000(0.8) = C_y(1.4) \qquad C_y = 1143$

Method of joint @ C $\qquad \sum F_y = 0$ $\qquad$ @ C



$F_{BC}\left(\frac{4}{5}\right) = 1143 \qquad F_{BC} = 1429$

$F_{AC} = F_{BC}\left(\frac{3}{5}\right) = 857$

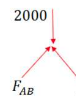
$F_{AB} = 1212\ N$

$F_{Bc} = 1429N$

$F_{AC} = 857\ N$

$P_{cr} = \frac{\pi^2 EI}{L^2}$

Method of joint @ B



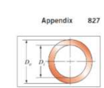
$\sum F_y = 0 \qquad F_{AB}\left(\frac{1}{\sqrt{2}}\right) = F_{BC}\left(\frac{3}{5}\right) \qquad F_{AB} = 1212\ N$

Column Analysis

2. The length of the post of the stool is 48". Assume *fixed-free* end condition. The mechanical tubing used is Ref. d (OD = 1 inch) as shown. Given the material has the properties of S_y = *50,000 psi* and *E = 10,000,000 psi*. Determine the safety factor *N* when a 300 lb person is sitting on the stool.

TABLE 15-18 Mechanical Tubing: Steel and Aluminum: 0.5 in to 5.0 in Outside Diameter

Ref.	Nominal size OD (in)	Nominal size Wall gauge	Outside diameter D_o (in)	Inside diameter D_i (in)	Wall thickness t_w (in)	Area A (in²)	Weight/ft[2] Steel (lb/ft)	Weight/ft[2] Aluminum (lb/ft)	Section properties[1] Bending I (in⁴)	Bending S (in³)	r (in)	Torsion J (in⁴)	Torsion Z_p (in³)
a	1/2	17	0.500	0.384	0.058	0.081	0.274	0.097	0.00200	0.00800	0.158	0.00400	0.0160
b	1/2	14	0.500	0.334	0.083	0.109	0.370	0.130	0.00246	0.00983	0.150	0.00491	0.0197
c	1	16	1.000	0.870	0.065	0.191	0.649	0.229	0.0210	0.0419	0.331	0.0419	0.0839
d	1	10	1.000	0.732	0.134	0.365	1.239	0.437	0.0350	0.0700	0.310	0.0700	0.140



End condition, Fixed-free, K = 2

$$S_r = \frac{L_{eff}}{r} = \frac{2(48)}{0.310} = 309$$

$$C_c = \sqrt{\frac{2\pi^2 E}{S_y}} = \sqrt{\frac{2\pi^2(10\times 10^6)}{50000}} = 62.83$$

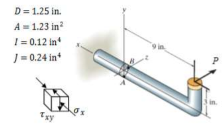
$S_r > C_c \rightarrow$ long column, Euler

$$P_{cr} = \frac{\pi^2 EI}{L_{eff}^2} = \frac{\pi^2(10\times 10^6)(0.035)}{96^2} = 374.8$$

$$N = \frac{374.8}{300} = 1.25$$

The force P repeats from 0 lb to 2000 lb. The material properties: $S_u = 120$ ksi and $S_n' = 30$ ksi. Calculate the stresses at location B (at the right). Determine the safety factor at location B using Goodman criterion.

$D = 1.25$ in.
$A = 1.23$ in^2
$I = 0.12$ in^4
$J = 0.24$ in^4



$F_x = 0$
$F_y = 0$
$F_z = 2000$
$M_x = 2000(3) = 6000$
$M_y = 2000(9) = 18000$
$M_z = 0$

shear: $\tau = 0$ — B is at the bottom edge, Q = 0

torsion: $\tau_{xy} = \frac{Tc}{J} = \frac{6000(0.625)}{0.24} = 15625$

bending: $\sigma_{xy} = \frac{Mc}{I} = \frac{18000(0.625)}{0.12} = 93750$ compression

$\sigma_{x,max} = 0$ $\sigma_{x,min} = -93750$ $\sigma_{y,max} = 0$ $\sigma_{y,min} = 0$ $\tau_{xy,max} = 15625$ $\tau_{xy,min} = 0$	$\sigma_{x,m} = -46875$ $\sigma_{x,a} = 46875$ $\sigma_{y,m} = 0$ $\sigma_{y,a} = 0$ $\tau_{xy,m} = 7812$ $\tau_{xy,a} = 7812$	**Mean** $\sigma_{x,m} = -46875$ $\sigma_{y,m} = 0$ $\tau_{xy,m} = 7812$ $\sigma_{m1} = 1267$ $\sigma_{m2} = 0$ $\sigma_{m3} = -48134$ MSST $\sigma_m' = -49410$	**Alternating** $\sigma_{x,a} = 46875$ $\sigma_{y,a} = 0$ $\tau_{xy,a} = 7812$ $\sigma_{a1} = 48134$ $\sigma_{a2} = 0$ $\sigma_{a3} = -1267$ MSST $\sigma_a' = 49410$

$\sigma_c = -23438$
$r = 24705$

$\sigma_c = 23438$
$r = 24705$

3. The post in Problem 2 was found to have an initial crookedness of 0.2 in. Determine the maximum allowable force it can carry.

It is a long column, and the Euler critical load P_{cr} is 374.8 lb

$A = 0.365$
$I = 0.0350$
$r = \sqrt{I/A} = 0.310$
$c = D/2 = 0.5$
$a = 0.2$

$$C_1 = -\frac{1}{N}\left[S_y A + \left(1 + \frac{ac}{r^2}\right)P_{cr}\right]$$

$$C_1 = -\frac{1}{1}\left[50000(0.365) + \left(1 + \frac{(0.2)(0.5)}{0.310^2}\right)375\right] = -19015$$

$$C_2 = \frac{S_y A P_{cr}}{N^2}$$

$$C_2 = \frac{50000(0.365)(375)}{1^2} = 6843750$$

$P_a^2 + C_1 P_a + C_2 = 0$

$P_a^2 - 19015 P_a + 6843750 = 0$

$P_a = 367$ ← Reduced from 375 lb

$P_a = 18648$

Buckling

4. If the post is straight, but a 300 lb person sits on the edge of the chair, at a distance 4.0 in. from the center line of the post. Determine the safety factor.

$P = 300$
$A = 0.365$
$e = 4.0$
$c = D/2 = 0.5$
$I = 0.0350$
$r = \sqrt{I/A} = 0.310$
$L_{eff} = 96$

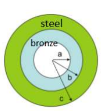
$$\sigma_{max} = \frac{P}{A}\left[1 + \frac{ec}{r^2}\sec\left(\frac{L_{eff}}{2r}\sqrt{\frac{P}{AE}}\right)\right]$$ compression

$$\sigma_{max} = \frac{300}{0.365}\left[1 + \frac{(4.0)(0.5)}{(0.310)^2}\sec\left(\frac{96}{2(0.310)}\sqrt{\frac{300}{(0.365)(10 \times 10^6)}}\right)\right] = 103707\ psi$$ compression

$$N = \frac{S_y}{\sigma_{max}} = \frac{50000}{103707} = 0.482$$

Interference Fits

1. A bronze bushing (inner member) is to be installed into a steel sleeve (outer member). The bushing has an ID of 2.0 in. and nominal OD of 2.4 in. The steel sleeve has a nominal ID of 2.4 in. and an OD of 2.8 in.
   a. With the nominal diameter of 2.4 in., specify the tolerance of the bushing and sleeve based on FN3 interference fit.
   b. Use the maximum interference to calculate the interface pressure.
   c. Calculate the stresses (axial, hoop and radial stresses) at the inner wall of the outer member (steel sleeve).
   d. Based on MSST, calculate the effective stress and determine the required strength based on N=2.



$a = 1.0, b = 1.2, c = 1.4$

$\nu_o = \nu_{steel} = 0.27$
$E_o = E_{steel} = 30 \times 10^6$

$\nu_i = \nu_{bronze} = 0.27$
$E_i = E_{bronze} = 17 \times 10^6$

a)

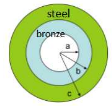
FN3 Fit
Hole: 2.4000 +0.0012 / -0.0000
Hole: 2.4000 +0.0032 / +0.0025
Limits of interference: 0.0012
0.0032 = δ

b)

$$p = \frac{\delta}{2b\left[\frac{1}{E_o}\left(\frac{c^2+b^2}{c^2-b^2} + \nu_o\right) + \frac{1}{E_i}\left(\frac{b^2+a^2}{b^2-a^2} - \nu_i\right)\right]} = \frac{0.0032}{2(1.2)\left[\frac{1}{30 \times 10^6}\left(\frac{3.4}{0.52} + 0.27\right) + \frac{1}{17 \times 10^6}\left(\frac{2.44}{0.44} - 0.27\right)\right]}$$

$$p = \frac{0.0032 \times 10^6}{2(1.2)(0.538)} = 2478\ psi$$

outer member: critical @ inner surface



$p_o = 0, p_i = p$

c)

$\sigma_a = 0 = \sigma_2$

$$\sigma_\theta = \frac{p(1.2^2)}{1.4^2 - 1.2^2}\left(1 + \frac{1.4^2}{1.2^2}\right) = 6.54(p) = 16206 = \sigma_1$$

$$\sigma_r = \frac{p(1.2^2)}{1.4^2 - 1.2^2}\left(1 - \frac{1.4^2}{1.2^2}\right) = -p = -2478 = \sigma_3$$

d)

MSST

$$\sigma' = \sigma_1 - \sigma_3 = 16206 + 2478 < \frac{S_y}{2}$$

$S_y > 37368\ psi$

2. Given the same bushing and sleeve as in the previous problem (A bronze bushing (inner member) is to be installed into a steel sleeve (outer member). The bushing has an ID of 2.0 in. and nominal OD of 2.4 in. The steel sleeve has a nominal ID of 2.4 in. and an OD of 2.8 in.), calculate the maximum interference (δ) allowed based on the yield strength of the outer steel member $S_y = 50\ ksi$ and safety factor N = 2.

$\sigma_a = 0 = \sigma_2$

$$\sigma_\theta = \frac{p(1.2^2)}{1.4^2 - 1.2^2}\left(1 + \frac{1.4^2}{1.2^2}\right) = 6.54(p) = \sigma_1$$

$$\sigma_r = \frac{p(1.2^2)}{1.4^2 - 1.2^2}\left(1 - \frac{1.4^2}{1.2^2}\right) = -p = \sigma_3$$

MSST

$$\sigma' = \sigma_1 - \sigma_3 = 6.54p - (-p) = 7.54p < \frac{S_y}{2}$$

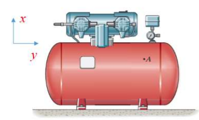
$p < \frac{50000}{2(7.54)}$ $p < 3315\ psi$

$$p = 3315 = \frac{\delta}{2b\left[\frac{1}{E_o}\left(\frac{c^2+b^2}{c^2-b^2} + \nu_o\right) + \frac{1}{E_i}\left(\frac{b^2+a^2}{b^2-a^2} - \nu_i\right)\right]} = \frac{\delta}{2(1.2)\left[\frac{1}{30 \times 10^6}\left(\frac{3.4}{0.52} + 0.27\right) + \frac{1}{17 \times 10^6}\left(\frac{2.44}{0.44} - 0.27\right)\right]}$$

$\delta = 0.00427\ in$

Since it is linearly proportional, a quicker way is to use the information in the previous answer (a) and (b) to calculate the interference

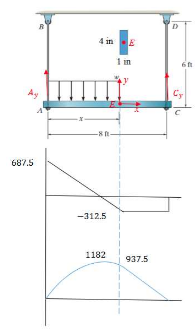
$\delta = 0.0032\ in$, $p = 2478\ psi$ ⇨ $\delta = ?\ in$, $p = 3315\ psi$ $\delta = \frac{3315}{2478}(0.0032) = 0.00428\ in$



The tank is subjected to a cyclic internal pressure of 20 psi to 80 psi. If the internal diameter of the tank is 20 in., the wall thickness is 0.1 in., the material properties are *Sy=60 ksi, Sn'=30 ksi*, use DET effective stress and Soderberg criterion to determine the safety factor of the design.

$\sigma_{x(Max)} = \frac{80 \times 10}{0.1} = 8000$ $\sigma_{x(Min)} = \frac{20 \times 10}{0.1} = 2000$	$\sigma_{x(Mean)} = \frac{8000 + 2000}{2} = 5000$ $\sigma_{x(Alternating)} = \frac{8000 - 2000}{2} = 3000$
$\sigma_{y(Max)} = \frac{80 \times 10}{2 \times 0.1} = 4000$ $\sigma_{y(Min)} = \frac{20 \times 10}{2 \times 0.1} = 1000$	$\sigma_{y(Mean)} = \frac{4000 + 1000}{2} = 2500$ $\sigma_{y(Alternmating)} = \frac{4000 - 1000}{2} = 1500$
$\tau_{xy(max)} = 0$ $\tau_{xy(min)} = 0$	$\tau_{xy(m)} = 0$ $\tau_{xy(a)} = 0$

The distributed load (x = 5 ft) repeats from w = 0 to 200 lb/ft. The material properties: $S_y = 75$ ksi and $S_n' = 25$ ksi. Given the cross section of beam is 4 in. × 1 in., calculate the stresses at location E (middle of the beam). Based on location E, determine the safety factor using Soderberg criterion.



@A $\sum M_z = 0$ $200(5)\left(\frac{5}{2}\right) = C_y(8)$ $C_{y,max} = 312.5$

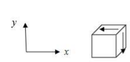
$\sum F_y = 0$ $A_y + 312.5 = (200)(5)$ $A_{y,max} = 687.5$

Construct shear and moment diagrams, at the cross-section of interest

$V_{y,max} = -312.5$ lb $M_{z,max} = 937.5$ lb.in

@E Neutral axis

$$\tau_{xy,max} = \frac{3V}{2A} = \frac{3(312.5)}{2(4)(1)} = 117\ \text{psi}$$



$\sigma_{x,max} = 0$ $\sigma_{x,min} = 0$ $\sigma_{y,max} = 0$ $\sigma_{y,min} = 0$ $\tau_{xy,max} = 118$ $\tau_{xy,min} = 0$	$\sigma_{x,m} = 0$ $\sigma_{x,a} = 0$ $\sigma_{y,m} = 0$ $\sigma_{y,a} = 0$ $\tau_{xy,m} = 59$ $\tau_{xy,a} = 59$	**Mean** $\sigma_{x,m} = 0$ $\sigma_{y,m} = 0$ $\tau_{xy,m} = 59$ $\sigma_{m1} = 59$ $\sigma_{m2} = 0$ $\sigma_{m3} = -59$ MSST $\sigma_m' = 118$	**Alternating** $\sigma_{x,a} = 0$ $\sigma_{y,a} = 0$ $\tau_{xy,a} = 59$ $\sigma_{a1} = 59$ $\sigma_{a2} = 0$ $\sigma_{a3} = -59$ MSST $\sigma_a' = 118$

$$\frac{K_f \sigma_a'}{S_n'} + \frac{\sigma_m'}{S_y} = \frac{1}{N}$$ $S_y = 75000$, $S_n' = 25000$

Soderberg

$$\frac{1.0(118)}{25000} + \frac{118}{75000} = \frac{1}{N}$$

N = 159